



Modelling Financial Markets

Chapter 15: LLMs and Sentiment Analysis



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- AI: Artificial Intelligence
- API: Application Programming Interface
- AUC: Area Under the (ROC) Curve
- BERT: Bidirectional Encoder Representations from Transformers
- BH: Benjamini-Hochberg (procedure)
- BY-NC: Creative Commons Attribution-NonCommercial
- CI: Confidence Interval
- FDR: False Discovery Rate
- FNSPID: Financial News and Stock Price Integration Dataset
- FWER: Family-Wise Error Rate
- GI: (Harvard) General Inquirer, dictionary IV-4
- GPT: Generative Pre-trained Transformer
- GPT-4: Generative Pre-trained Transformer 4
- GPU: Graphics Processing Unit
- IDA: Institute for Digital Assets
- IV-4: Harvard IV-4 (version 4 of the General Inquirer word lists)
- LLM: Large Language Model
- LM: Loughran-McDonald (financial sentiment dictionary)
- LR: Likelihood Ratio (test)
- MIT: Massachusetts Institute of Technology
- PCA: Principal Component Analysis
- RLHF: Reinforcement Learning from Human Feedback
- TF-IDF: Term Frequency - Inverse Document Frequency
- US: United States
- UTC: Coordinated Universal Time (Temps Universel Coordonné)
- VADER: Valence Aware Dictionary and sEntiment Reasoner

Outline

Why Text?

Dictionaries

Labelled Data and Evaluation

Fine-Tuned Language Models: FinBERT

Large Language Models

Embeddings as Features

Look-Ahead Bias and Memorisation

From Sentiment to a Trading Signal

AI Agents in Trading

AI for Scientific Discovery

Practice and Conclusions

References

Markets Run on Words

- Prices move when information arrives, and most information arrives as text
 - ▶ News, company filings, earnings calls, analyst reports, central-bank statements, social media
- Financial newspapers are as old as organised markets
 - ▶ The Wall Street Journal: first issue on 8 July 1889 (photo)
- Today: millions of documents per day, far more than any analyst can read



The Wall Street Journal, first issue, 8 July 1889

Photo: Dow Jones & Company (1889); public domain;
[Wikimedia Commons](#)

Text as Data

- Gentzkow, Kelly & Taddy (2019): a text becomes a numerical array, which feeds an ordinary statistical model
 - ▶ Step 1: represent the document d as numbers x_d (counts, scores, vectors)
 - ▶ Step 2: map x_d to a quantity of interest (sentiment, topic, risk)
 - ▶ Step 3: use that quantity in an economic question (returns, volatility, decisions)
- Surveys for finance: Kearney & Liu (2014), Loughran & McDonald (2016)
- This chapter: step 2 for sentiment, then step 3 for daily stock returns

What We Mean by Sentiment

- ▣ **Sentiment of a text:** is it good, bad or neutral news for the asset it mentions?
 - ▶ Three classes: positive, neutral, negative; or a continuous score in $[-1, 1]$
 - ▶ Net score from class probabilities: $s = P(\text{positive}) - P(\text{negative})$
- ▣ Not the same as **investor sentiment**: the mood of investors, measured from market data ([Baker & Wurgler \(2006\)](#))
 - ▶ Text sentiment is one input into investor sentiment, not a synonym for it

Early Evidence: Words Move Prices

- ▣ [Antweiler & Frank \(2004\)](#): internet message-board posts about stocks help predict volatility; the effect on returns is small
- ▣ [Tetlock \(2007\)](#): high media pessimism in a Wall Street Journal column predicts downward pressure on prices, followed by a reversal
 - ▶ Unusually high or low pessimism also predicts high trading volume
- ▣ [Tetlock, Saar-Tsechansky & Macskassy \(2008\)](#): the share of negative words in firm-specific news predicts low earnings and returns that are only slowly absorbed
- ▣ Common thread: text carries information that prices absorb, quickly but not always instantly

Narratives and Markets

- ▣ Shiller (2017): popular stories spread like epidemics and move economic decisions
 - ▶ Narratives can be measured only through text: newspapers, books, social media
- ▣ Shiller: Nobel Prize in Economics 2013, with Fama and Hansen
 - ▶ Chapter 2 (efficiency) and Chapter 17 (bubbles) meet here: narratives can push prices away from fundamentals



Robert J. Shiller, Stockholm, December 2013

Photo: [Bengt Nyman \(2013\)](#); CC BY 2.0; [Wikimedia Commons](#)

Four Ways to Turn Text into Sentiment

Method	How it works	Labels needed	Examples
Dictionary	count listed words	none	GI, LM, VADER
Fine-tuned model	network trained on labelled sentences	thousands	FinBERT
Large language model	instruction in plain language (prompt)	none (zero-shot)	Qwen2.5, GPT
Embeddings + classifier	text vectors into a regression	hundreds to thousands	MiniLM

- We compare all four on the same labelled data, with confidence intervals
- Then we ask whether any of them makes money on daily stock returns

Section Summary

- ▣ Most financial information is text; text becomes data through a numerical representation
- ▣ Text sentiment: the tone of a document about an asset, as classes or as a score in $[-1, 1]$
- ▣ Four families of methods, from word lists to large language models

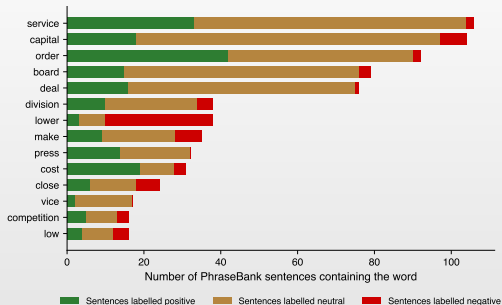
Bag of Words and Tone

- **Bag of words:** a document is reduced to its word counts; order is ignored
 - ▶ Tokens: words in capital letters, without digits and punctuation
- A dictionary lists positive words $\mathcal{P}$ and negative words $\mathcal{N}$
 - ▶ P_d, N_d : number of positive and negative words in document d
 - ▶ **Tone:** $\tau_d = \frac{P_d - N_d}{P_d + N_d} \in [-1, 1]$, and $\tau_d = 0$ if $P_d + N_d = 0$
 - ▶ **Class:** positive if $\tau_d > 0$, negative if $\tau_d < 0$, neutral otherwise
- Transparent, fast and free of training data; blind to context and negation

Generic versus Financial Dictionaries

- **Harvard General Inquirer** (GI), list IV-4: a psychology dictionary from the 1960s
 - ▶ 2 007 negative and 1 637 positive words
- **Loughran & McDonald (2011)** (LM): built from annual reports (10-K filings) of US firms
 - ▶ 2 355 negative and only 354 positive words; also lists for uncertainty and litigation
 - ▶ Their finding: almost three quarters of the GI negative words are not negative in a financial context
- In the lists used here: 1 552 GI negative words (77%) are not LM negative words

Negative for a Psychologist, Neutral for an Accountant



- Words negative in GI but not in LM, ranked by the number of Financial PhraseBank sentences that contain them
- Colour: the sentiment label that human annotators gave to the sentence

Interpretation: Finance Has Its Own Vocabulary

- "Service" (106 sentences), "capital" (104) and "board" (79) are business nouns, not bad news
 - ▶ Share of their sentences not labelled negative: 98%, 93%, 96%
 - ▶ "Vice" appears in "vice president": 100% of its sentences are not negative
- "Lower" is the exception: only 26% not negative, but "lower costs" is good news
 - ▶ The sign depends on the next word: a problem no word list can solve
- Of the 964 sentences whose only negative words are of this kind, 89% are not labelled negative

Worked Example: One Sentence, Two Dictionaries

- Financial PhraseBank sentence, labelled **positive** by all annotators:
 - ▶ *Uponor improved its performance in spite of the decrease in residential construction in the US .*
- LM: positive words improved; negative words –
 - ▶ $\tau = (1 - 0)/(1 + 0) = +1.00$: positive, correct
- GI: positive words –; negative words spite, decrease
 - ▶ $\tau = (0 - 2)/(0 + 2) = -1.00$: negative, wrong
 - ▶ GI misses "improved" and counts "spite" and "decrease", which describe the market, not the firm

VADER: Rules on Top of a Word List

- [Hutto & Gilbert \(2014\)](#): a lexicon with intensity scores for about 7,500 terms, built for social media
 - ▶ Rules for negation ("not good"), intensifiers ("very"), capital letters, exclamation marks and emoticons
 - ▶ Output: a compound score in $[-1, 1]$; usual cut-offs ± 0.05 for positive and negative
- Rules fix some context problems, but the vocabulary is still generic, not financial
 - ▶ It is the second generic benchmark in our tests, next to GI

Section Summary

- Dictionary tone: $(P - N)/(P + N)$ from counts of listed words
- Generic dictionaries misread finance: most GI negative words are ordinary business words
- LM fixes the vocabulary, not the context: "lower costs" still counts as negative

Financial PhraseBank

- Malo et al. (2014): 4 846 sentences from financial news about listed firms
 - ▶ Each sentence labelled by 16 annotators with background in finance and business
 - ▶ Question: is the sentence good, bad or neutral news for the investor?
- Agreement levels: at least 50%, 66%, 75% or 100% of annotators chose the label
 - ▶ 2 264 sentences with full agreement; 629 where only half agreed
- Labels: 59% neutral, 28% positive, 12% negative

Twitter Financial News

- Headlines of financial news published on Twitter in 2022, labelled bearish, bullish or neutral
 - ▶ Training set of 9,543 and validation set of 2 388 headlines; we evaluate on the validation set
 - ▶ Labels: 66% neutral, 20% positive, 15% negative
- Short, noisy texts with tickers and links: closer to the real news flow than PhraseBank
 - ▶ Published under the MIT licence on the Hugging Face Hub
- Two data sets, because a result on one set may not carry over to another

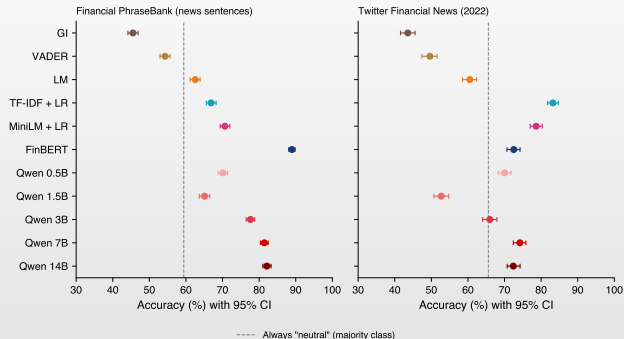
Measuring a Classifier

- **Accuracy:** share of texts with the correct label
 - ▶ Benchmark: always predict "neutral" gives 66% on Twitter and 59% on PhraseBank
- For class k : $\text{precision} = \frac{TP_k}{TP_k + FP_k}$, $\text{recall} = \frac{TP_k}{TP_k + FN_k}$, $F1_k = \frac{2 \text{prec} \cdot \text{rec}}{\text{prec} + \text{rec}}$
 - ▶ TP_k : texts of class k labelled k ; FP_k : other texts labelled k ; FN_k : texts of class k labelled otherwise
- **Macro-F1:** average of $F1_k$ over the three classes; each class counts equally
 - ▶ "Always neutral" gets macro-F1 of only 26.4% on Twitter

Inference for Classifiers

- Bootstrap interval: resample the n texts with replacement, recompute accuracy, take the 2.5% and 97.5% quantiles
 - ▶ With $n \approx 2,400$ texts and accuracy near 70%, the 95% CI is about ± 1.8 percentage points
- **McNemar test**: two classifiers on the same texts
 - ▶ n_{01} : A wrong and B right; n_{10} : A right and B wrong; texts where both agree carry no information
 - ▶ Under H_0 (equal accuracy), $n_{01} \sim \text{Binomial}(n_{01} + n_{10}, 0.5)$: an exact test
- Paired tests are far more powerful than comparing two separate intervals

Eleven Classifiers, Two Data Sets

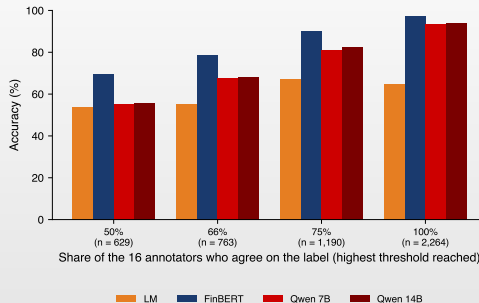


- TF-IDF + LR and MiniLM + LR: logistic regressions trained on the Twitter training set; all other methods use no Twitter labels

Interpretation: No Single Winner

- ▣ Generic dictionaries lose to the majority class on both sets: GI 43.6%, VADER 49.6% on Twitter
 - ▶ LM is better (60.5%) but still below "always neutral" (66%) on Twitter
- ▣ FinBERT: 89.0% on PhraseBank, 72.5% on Twitter; Qwen2.5-7B zero-shot: 81.4% and 74.2%
 - ▶ The best on Twitter is the simplest supervised model: TF-IDF + logistic regression, 83.2%
 - ▶ TF-IDF: counts of words and word pairs, weighted down when a term appears in many documents
- ▣ Lesson: labelled data from the target domain beat model size

When Humans Disagree, Models Disagree Too



- Financial PhraseBank split by the highest agreement level reached by the 16 annotators

Interpretation: Ambiguity Sets a Ceiling

- ▣ Full agreement: FinBERT 97%, Qwen2.5-14B 94%, LM 65%
 - ▶ Only half of the annotators agree: FinBERT 70%, Qwen2.5-14B 55%, LM 54%
- ▣ Many sentences are genuinely ambiguous: even experts split
 - ▶ A model cannot be judged on a label that humans do not agree on
- ▣ For signals: a strong, unambiguous tone is more informative than a borderline one

Section Summary

- ▣ Two labelled data sets: Financial PhraseBank (news sentences) and Twitter Financial News (headlines)
- ▣ Report accuracy with a CI and macro-F1; compare classifiers with the paired McNemar test
- ▣ Generic dictionaries fail; supervised models from the target domain are hard to beat

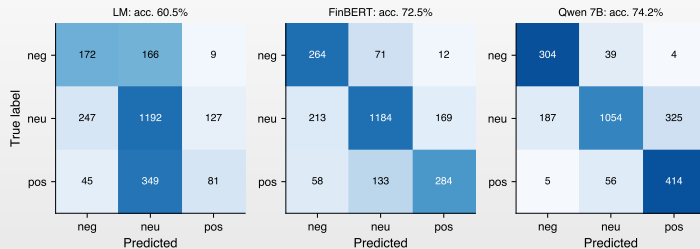
From Words to Words in Context

- A dictionary gives "lower" one fixed sign; a language model reads the sentence around it
 - ▶ Each token gets a vector that depends on all other tokens (attention, Chapter 14; [Vaswani et al. \(2017\)](#))
- **BERT** ([Devlin et al. \(2019\)](#)): a Transformer encoder pretrained on books and Wikipedia
 - ▶ Pretraining task: guess words hidden at random in a sentence (masked language modelling)
 - ▶ About 110 million parameters in the base version
- Fine-tuning: add a small classification layer and train the whole network on labelled sentences

FinBERT

- [Araci \(2019\)](#): BERT further pretrained on financial news, then fine-tuned for sentiment
 - ▶ The public model used here (ProsusAI/finbert) was fine-tuned on Financial PhraseBank
 - ▶ Output: probabilities of positive, negative and neutral
- [Huang, Wang & Yang \(2023\)](#): a FinBERT trained on analyst reports, earnings-call transcripts and 10-K filings
 - ▶ Outperforms dictionaries and other machine-learning methods, mainly on sentences that dictionaries misread
- Runs on an ordinary laptop: a few milliseconds per sentence

Where the Errors Are



□ Twitter validation set: rows are the true labels, columns the predicted labels; counts in the cells

Interpretation: Different Models, Different Mistakes

- ▣ LM sends most positive headlines to neutral: its positive list is short (354 words)
- ▣ FinBERT errs in both directions around neutral
- ▣ Qwen2.5-7B finds almost all positive and negative headlines, but calls many neutral ones positive or negative
 - ▶ An LLM "sees" sentiment where annotators saw plain facts: high recall, low precision on the extreme classes
- ▣ For trading, false positives cost money: precision on the extreme classes matters

The In-Sample Trap

- ▣ FinBERT: 89.0% on PhraseBank, the data it was trained on; 72.5% on Twitter
 - ▶ Drop: about 16 percentage points
- ▣ Qwen2.5-7B, never trained on these labels: 81.4% and 74.2%
 - ▶ Part of FinBERT's lead on PhraseBank is memory of its own training data
- ▣ Rule: evaluate a language model only on texts it has never seen
 - ▶ Same logic as out-of-sample testing in Chapters 8 and 13

Worked Example: Is FinBERT Better than LM?

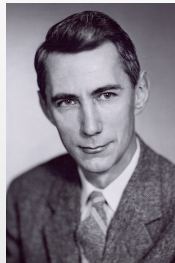
- ▣ Twitter validation set, 2 388 headlines: LM 60.5%, FinBERT 72.5%
 - ▶ LM wrong and FinBERT right: $n_{01} = 539$; LM right and FinBERT wrong: $n_{10} = 252$
 - ▶ Exact McNemar $p < 0.001$
- ▣ Difference +12.0 percentage points, bootstrap CI 95% [+10.0; +14.2]
 - ▶ Clear win for the fine-tuned model, even outside its training data

Section Summary

- ▣ Pretrained Transformers read words in context; fine-tuning adapts them with labelled sentences
- ▣ FinBERT beats every dictionary out of sample, but its PhraseBank score is inflated by training overlap
- ▣ Look at the confusion matrix, not only at accuracy

The Next-Word Problem

- Shannon (1951): how well can one guess the next letter of English text from the previous ones?
 - ▶ Language is highly predictable: its entropy is far below that of random letters
- A large language model (LLM) does the same with tokens:
 $P(w_{t+1} \mid w_1, \dots, w_t)$
 - ▶ Trained on trillions of tokens of text and code
 - ▶ Quantinar: The Next Word Problem and Quantinar: Natural Language Processing cover the background



Claude E. Shannon, founder of information theory

Photo: Tekniska museet, Stockholm (undated); CC

BY 2.0; Wikimedia Commons

From Prediction to Instructions

- ▣ [Brown et al. \(2020\)](#): a large enough next-word model solves tasks from a few examples in the prompt (few-shot)
- ▣ [Ouyang et al. \(2022\)](#): instruction tuning and RLHF teach the model to follow requests
 - ▶ Humans rank candidate answers; the model is trained to produce highly ranked ones
- ▣ Finance-specific LLMs: BloombergGPT ([Wu et al. \(2023\)](#)), 50 billion parameters, trained on financial and general text
 - ▶ General open models now match many finance-specific ones on sentiment tasks

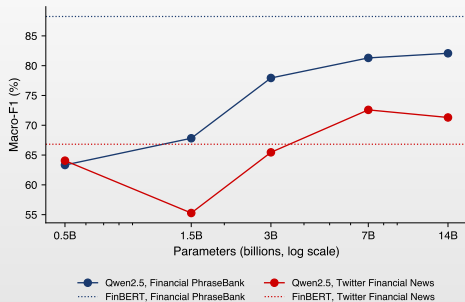
Zero-Shot Classification with an LLM

- Prompt: a system instruction plus the text to classify
 - ▶ You classify the sentiment of financial news for investors. Answer with one word: positive, negative or neutral.
 - ▶ Headline: <text> Sentiment:
- We read the probabilities of the three answer words at the first generated position
 - ▶ Logits z_k of the tokens "positive", "negative", "neutral": $p_k = e^{z_k} / \sum_j e^{z_j}$
 - ▶ Label: $\arg \max_k p_k$; score: $p_{\text{pos}} - p_{\text{neg}}$; no text is generated, so nothing needs parsing

The Models: Qwen2.5-Instruct

- ▣ [Qwen Team \(2024\)](#): open-weight models from 0.5 to 72 billion parameters, weights published on 19 September 2024
 - ▶ We use 0.5B, 1.5B, 3B, 7B and 14B (B: billion parameters), all under open licences
- ▣ Speed on a laptop graphics processor (GPU), per headline, including prompt processing
 - ▶ 0.5B: 4 ms; 1.5B: 12 ms; 3B: 20 ms; 7B: 92 ms; 14B: 88 ms
 - ▶ Open weights: results can be reproduced, and the release date bounds what the model could know

Bigger Is Not Always Better



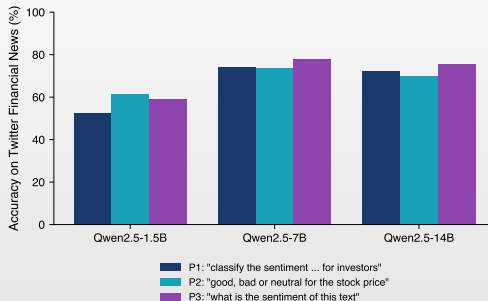
Macro-F1 of zero-shot Qwen2.5 with the same prompt; dotted lines: FinBERT

 MFM_ch15_llm_zero_shot

Interpretation: Size Helps, Unevenly

- ▣ PhraseBank: macro-F1 rises from 63.3% (0.5B) to 82.1% (14B), still below FinBERT (88.3%)
 - ▶ FinBERT's lead there is partly training overlap (previous section)
- ▣ Twitter: not monotone; 1.5B is the worst (55.3%), 7B the best (72.6%), above FinBERT (66.8%)
 - ▶ 7B against 1.5B on accuracy: +21.5 points, McNemar $p < 0.001$
- ▣ Small models react strongly to wording: the 1.5B model rarely answers "neutral" with this prompt

Same Model, Different Wording



- Three prompts that ask the same question; Twitter validation set

Interpretation: the Prompt Is a Parameter

- ▣ Accuracy ranges: 1.5B from 52.7% to 61.7%; 7B from 73.7% to 77.8%; 14B from 69.9% to 75.6%
 - ▶ The three prompts give the same label on only 67% of headlines (7B)
 - ▶ Share of "neutral" answers for 7B: 48% (P1) to 69% (P3)
- ▣ Choosing the prompt after seeing test results is a form of data snooping (Chapter 13)
 - ▶ Fix the prompt in advance, or choose it on a separate validation set

Worked Example: LLM versus FinBERT

- ▣ Twitter, prompt P1: Qwen2.5-7B 74.2%, FinBERT 72.5%
 - ▶ $n_{01} = 395$, $n_{10} = 355$; McNemar $p = 0.154$; difference +1.7 points, CI 95% [-0.5; +3.8]
 - ▶ Qwen2.5-14B against FinBERT: -0.1 points, $p = 0.942$
- ▣ No significant difference: a model with 70 to 130 times more parameters, several times slower, is not more accurate
 - ▶ Where LLMs help: no labels needed, any language, longer and more complex documents

Beyond Headlines: Long Documents

- ▣ Annual reports: [Loughran & McDonald \(2011\)](#) built their lists from 10-K filings; tone and uncertainty predict returns and volatility around filings
- ▣ Earnings calls and analyst reports: [Huang, Wang & Yang \(2023\)](#) show FinBERT extracts information that dictionaries miss
- ▣ [Kim, Muhn & Nikolaev \(2023\)](#): LLM summaries of disclosures are much shorter but explain stock-market reactions better than the originals
 - ▶ They measure disclosure "bloat": long, uninformative text is linked to lower price efficiency
- ▣ Long documents need models with long context windows, or chunking and aggregation

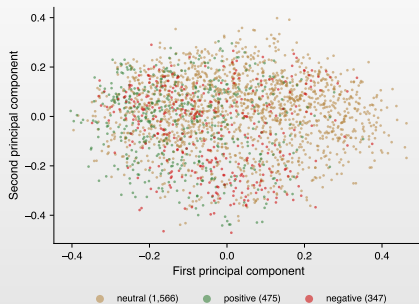
Section Summary

- ▣ An LLM predicts the next token; zero-shot sentiment reads the probabilities of the answer words
- ▣ Accuracy grows with size on average, but not monotonically; the prompt changes the result
- ▣ On short financial headlines, a 7B to 14B open model ties with FinBERT

Text Embeddings

- **Embedding:** a vector $e_d \in \mathbb{R}^k$ such that similar texts get nearby vectors
 - ▶ Word vectors: [Mikolov et al. \(2013\)](#) (word2vec) learn them from which words appear near each other
 - ▶ Sentence vectors: [Reimers & Gurevych \(2019\)](#) (Sentence-BERT) train a Transformer so that paraphrases are close
- Here: MiniLM ([Wang et al. \(2020\)](#)), 22 million parameters, $k = 384$; mean of token vectors, scaled to length 1
 - ▶ Similarity: $\cos(e_a, e_b) = e_a^\top e_b$ for unit vectors
- Use: embeddings enter an ordinary classifier or regression, like any other feature

What Embeddings Look Like

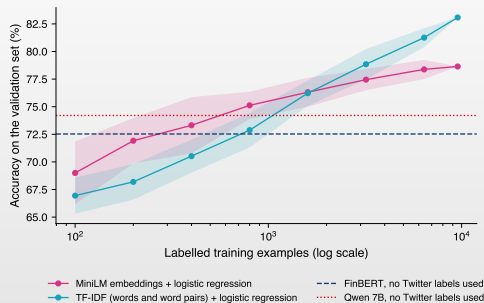


- Twitter validation headlines: first two principal components of the 384-dimensional MiniLM vectors, coloured by label

Interpretation: Sentiment Is Not the Main Direction

- The first two components separate topics (companies, macro news, crypto), not sentiment
 - ▶ PCA (Chapter 3) keeps the directions of largest variance, which need not be the ones we care about
- A supervised model finds the sentiment direction among the 384 dimensions
 - ▶ Logistic regression on all dimensions: 78.6% accuracy on Twitter

How Many Labels Do You Need?



- Logistic regression trained on random subsets of the Twitter training set, 10 repetitions per size; band: ± 2 standard deviations

Interpretation: a Few Hundred Labels Beat FinBERT

- ▣ Embeddings + logistic regression: 69.0% with 100 labels, above FinBERT from 400 labels
 - ▶ The pretrained vectors already contain most of what is needed
- ▣ TF-IDF starts lower (66.9%) but keeps improving: 83.1% with all 9,543 labels, against 78.6% for embeddings
 - ▶ With many labels, exact words ("upgrade", "cuts", "misses") carry the signal in headlines
- ▣ Practical rule: label a few hundred documents from your own domain before trusting any off-the-shelf model

Text Features in Return Prediction

- ▣ [Jegadeesh & Wu \(2013\)](#): word weights estimated from market reactions to 10-K filings, instead of fixed lists
- ▣ [Ke, Kelly & Xiu \(2019\)](#): a supervised model that learns sentiment words from news and the returns that follow
 - ▶ Their text-based signal predicts returns, most strongly on the day of and the day after the news
- ▣ [Gu, Kelly & Xiu \(2020\)](#) (Chapter 13): machine learning on many characteristics; text features can join the same framework
- ▣ Common rule: estimate the text model only on data from before the forecast period

Section Summary

- ▣ Embeddings turn texts into vectors that enter ordinary regressions
- ▣ Sentiment is one direction among many; supervision finds it
- ▣ A few hundred in-domain labels beat a generic fine-tuned model; many labels favour simple word features

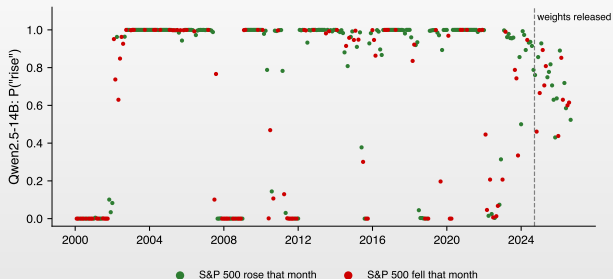
The Model May Know the Answer

- An LLM trained on text up to 2024 has read what happened after every 2015 headline
 - ▶ A backtest on 2015 news then mixes forecasting with remembering
- Glasserman & Lin (2023): two biases in sentiment backtests
 - ▶ Look-ahead: the model knows the returns that followed the news
 - ▶ Distraction: general knowledge about the named company interferes with reading the text; in their tests it mattered more than look-ahead
- Sarkar & Vafa (2024): direct tests show look-ahead in LLM applications to earnings calls and elections

Our Test: Ask About the Past, Without Context

- Monthly: "Did the S&P 500 index rise or fall during <month year>?", every month from 296 pre-release months plus 23 after
 - ▶ Daily: "Did Apple stock close higher or lower on <date> than on the previous trading day?", 500 random days 2010–2023 and 493 days after October 2024
- Score: $P(\text{rise})$ from the answer-word probabilities; true direction from the saved price series
 - ▶ AUC: probability that a random up period gets a higher score than a random down period; 0.5 = no knowledge (Chapter 13)
- After the release of the weights, the model cannot know: AUC must be 0.5 there

What Qwen2.5-14B "Remembers" About the S&P 500

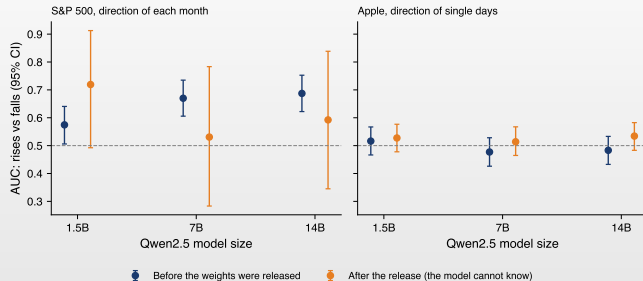


- Probability of the answer "rise" for each month, no context given; colour: what the index actually did

Interpretation: It Remembers Regimes, and Some Famous Months

- "Fall" for 2000–2002 and 2008–2009, "rise" for most of 2003–2007 and 2013–2021: the bear and bull markets
 - ▶ October 2008 (-16.9%), March 2020 (-12.5%), December 2018 (-9.2%): $P(\text{rise}) \approx 0$
 - ▶ But September 2002 (-11.0%) and February 2009 (-11.0%) get $P(\text{rise})$ close to 1
- After the release the answers drift towards "rise": no knowledge, only the general belief that markets go up
- Knowledge is coarse: memorable episodes, not every month

Memory Measured: AUC Before and After Release



□ Bootstrap 95% intervals; 0.5 means the model cannot tell up from down

Interpretation: Months Yes, Days No

- ▣ Months before release: AUC 0.67 (7B, CI [0.61; 0.73]) and 0.69 (14B, CI [0.62; 0.75])
 - ▶ Permutation p -value for 14B: 0.000; the 1.5B model knows much less (0.57)
 - ▶ After release (23 months): intervals include 0.5, but are too wide to prove anything
- ▣ Single days of Apple: AUC between 0.48 and 0.52 before release; no daily memory
 - ▶ Look-ahead is a danger for backtests at the horizon of well-known events, less for daily noise

Remedies

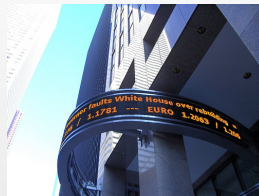
- Test only after the model's training cutoff: [Lopez-Lira & Tang \(2026\)](#) evaluate GPT-4 on post-knowledge-cutoff headlines
 - ▶ Clean, but the test window starts late and grows slowly
- Remove company names and dates from the text before scoring it ([Glasserman & Lin \(2023\)](#))
- Use models whose training data end before the test period (point-in-time models; [Sarkar & Vafa \(2024\)](#))
- Report the model version and the release date of its weights with every result

Section Summary

- ▣ LLMs trained on recent text carry knowledge of the past that a backtest must not use
- ▣ Qwen2.5-7B and 14B partly "know" the monthly direction of the S&P 500 before 2024, not single days
- ▣ Clean evidence needs post-cutoff data, anonymised text or point-in-time models

Data: Dated Headlines and Prices

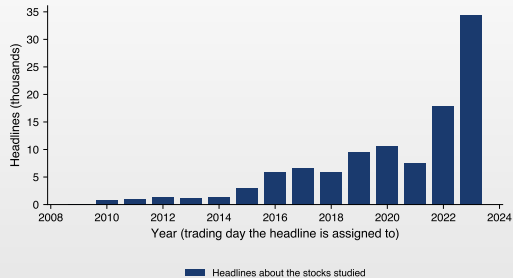
- ▣ FNSPID (Dong, Fan & Peng (2024)): 15.7 million news records about S&P 500 firms, 1999–2023, from four news websites
 - ▶ Licence CC BY-NC 4.0 (non-commercial use)
- ▣ Here: 16 large US stocks with daily prices in the course data (technology, banks, GameStop, Strategy)
 - ▶ 106 820 distinct headlines after removing repeats; 18 282 stock-days with news, 27.10.2009 to 18.12.2023



News ticker on the Reuters building, New York

Photo: bgilliard (2005); CC BY-SA 2.0; Wikimedia Commons

How Many Headlines, and When



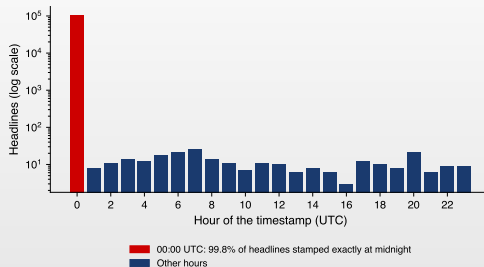
- Headlines about the 16 stocks per year, by the trading day they are assigned to

 MFM_ch15_news_data

Interpretation: Coverage Is Uneven

- 34 380 headlines in 2023 alone, against 8 445 for all years before 2016
 - ▶ Results are dominated by recent years; early years have few news days per stock
- On a news day a stock has on average 5.8 headlines
 - ▶ Daily score: the average of the headline scores for that stock and day
- Survivorship: the 16 stocks are today's large firms, chosen with hindsight

The Timestamp Problem



- Hour of the FNSPID timestamp in Coordinated Universal Time (UTC), log scale

 MFM_ch15_news_data

Interpretation: Dates, Not Times

- ▣ 99.8% of headlines are stamped exactly 00:00 UTC: the source gives the date, not the time of publication
 - ▶ A headline dated d may have appeared before the open, during the session or after the close of day d
- ▣ Alignment used here, with no look-ahead
 - ▶ Day d : the first trading day on or after the date of the headline
 - ▶ Returns on d and $d + 1$ may still contain the reaction; the first safe forecast is the return on $d + 2$ (close of $d + 1$ to close of $d + 2$)
- ▣ Excess return: stock return minus the return of SPY (an S&P 500 index fund) on the same day

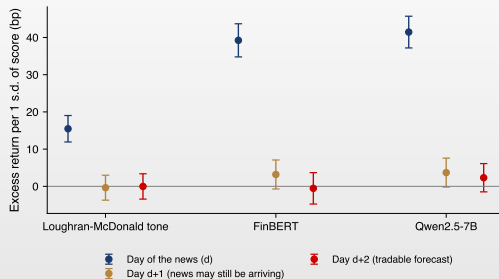
Three Scores for Every Headline

- LM tone; FinBERT net score $P(\text{pos}) - P(\text{neg})$; Qwen2.5-7B net score with prompt P1
 - ▶ LM finds a listed word in only 27% of headlines: the rest get tone 0
 - ▶ FinBERT labels: 19% negative, 62% neutral, 18% positive
- Correlations across headlines: FinBERT–Qwen 0.62, FinBERT–LM 0.41, Qwen–LM 0.36
 - ▶ The two language models agree much more with each other than with the dictionary
- Scores are standardised over the panel (mean 0, standard deviation 1) before regressions

Regression Design

- Pooled regression over stock-days with news: $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon_{i,d+h}$, $h = 0, 1, 2$
 - ▶ r^e : excess return (%); z : standardised daily score; b : return per one standard deviation of score
- Errors of stocks on the same day are correlated (common shocks): cluster standard errors by day (Petersen (2009))
 - ▶ $\hat{V} = (X^\top X)^{-1} \left(\sum_g X_g^\top \hat{\varepsilon}_g \hat{\varepsilon}_g^\top X_g \right) (X^\top X)^{-1}$, g : trading days
- $h = 0$ measures reaction, $h = 2$ measures predictability

Reaction versus Prediction



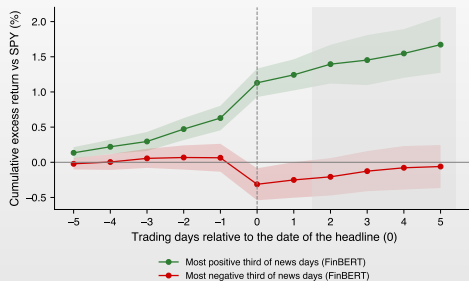
□ Slope b in basis points (bp) per one standard deviation of score, with 95% CI clustered by day

 MFM_ch15_news_signal

Interpretation: Headlines Describe Prices

- ▣ Day d : FinBERT +39.2 bp ($t = +17.27$), Qwen +41.4 bp ($t = +19.07$), LM +15.5 bp ($t = +8.53$)
 - ▶ Many headlines report the price move itself ("shares jump", "stock falls"): causality runs both ways
- ▣ Day $d + 1$: FinBERT +3.2 bp ($t = +1.60$), Qwen +3.7 bp ($t = +1.86$)
 - ▶ Part of this is news published after the close of d , not a forecast
- ▣ Day $d + 2$: FinBERT -0.5 bp ($t = -0.25$), Qwen +2.3 bp ($t = +1.19$), LM -0.0 bp ($t = -0.01$)

Event Study around News Days



- Average cumulative excess return from day -5 to $+5$; news days split by tertiles of the daily FinBERT score; shaded: the tradable days $d + 2$ to $d + 5$

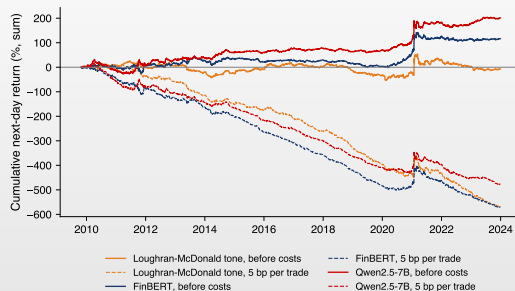
Interpretation: the Move Happens Before and On the News

- ▣ Most positive third: +0.63% over the 5 days before, +0.50% on day d , +0.11% on $d + 1$
 - ▶ Most negative third: +0.06%, -0.38% and +0.06%
- ▣ Tradable days $d + 2$ to $d + 5$: +0.43% (positive) and +0.19% (negative)
 - ▶ By the time a trader can act on the headline, the price has mostly adjusted: efficient markets (Chapter 2)
- ▣ Pre-event moves: information leaks, or headlines written after the price has already moved

A Daily Sentiment Strategy

- At the close of $d + 1$: for every stock with news on d , buy if its score is positive and sell short if negative
 - ▶ Each position hedged with SPY; equal weights; close after one day and earn the excess return of $d + 2$
 - ▶ Days with no news: no position, return 0
- Costs: 5 bp per trade; four trades per day with positions (stock and SPY, in and out)
 - ▶ Break-even cost: mean gross return on days with positions divided by 4
- Inference: Newey–West t (Chapter 2) and block-bootstrap CI for the mean daily return

Cumulative Strategy Returns



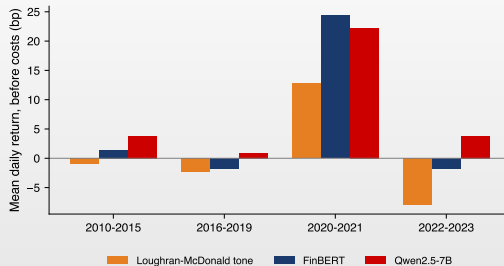
□ Sum of daily excess returns (%), before costs and with 5 bp per trade

 MFM_ch15_news_signal

Interpretation: a Small Signal, No Net Profit

- ▣ FinBERT: mean +3.27 bp per day ($t = +1.27$), CI 95% [-1.51; +8.93]: not different from zero
 - ▶ LM: -0.21 bp ($t = -0.07$)
- ▣ Qwen2.5-7B: +5.64 bp ($t = +2.26$), CI 95% [+0.95; +10.61]: significant before costs
 - ▶ Break-even cost +1.5 bp per trade, well below 5 bp: Sharpe +0.63 before and -1.48 after costs
 - ▶ Qwen2.5 was trained after 2023: part of its edge may be memorised prices (see Look-Ahead Bias and Memorisation)
- ▣ Non-tradable day $d + 1$: FinBERT -0.22 bp ($t = -0.10$), Qwen +4.06 bp ($t = +1.36$)
 - ▶ No significant gain even there: almost all of the reaction happens on day d

Stability over Time



- Mean daily return of the strategy before costs, by subperiod

 MFM_ch15_news_signal

Interpretation: Weak Evidence, Stated Honestly

- ▣ FinBERT by subperiod: 2010–2015 +1.4 bp, 2016–2019 -1.8 bp, 2020–2021 +24.3 bp, 2022–2023 -1.8 bp
 - ▶ Only 2020–2021 is large (FinBERT $t = +1.89$, Qwen +22.2 bp, $t = +1.78$): not significant at 5%, two years of meme-stock trading
 - ▶ Four trades of 5 bp cost 20 bp per day: even that period would barely break even
- ▣ Why so little: 16 heavily followed stocks, date-only timestamps, daily frequency
 - ▶ Published evidence of predictability relies on precise timestamps and many small stocks
- ▣ A negative result is still a result: it tells you where the signal is not

What the Literature Finds

- [Lopez-Lira & Tang \(2026\)](#): on headlines published after GPT-4's knowledge cutoff, GPT-4 captures the initial market reaction with portfolio-day hit rates of about 90%
 - ▶ That initial reaction is not tradable; GPT-4 scores also predict the later drift, most for small stocks and negative news
 - ▶ Strategy returns decline as LLM adoption rises: prices become more efficient
- [Bollen, Mao & Zeng \(2011\)](#): collective Twitter mood predicted the Dow Jones in 2008; the result is sensitive to the sample and method

Social Media and Crowds: GameStop, 2021

- January 2021: retail investors coordinated on social media and pushed GameStop up by more than ten times in a few weeks
 - ▶ Sentiment did not predict the move; the posts were the move
- Lessons for text signals
 - ▶ Volume of attention can matter more than tone
 - ▶ Crowded signals revert and create risk for short sellers (Chapter 17: bubbles)



A GameStop store, Vallejo, California, February 2021

Photo: Will Buckner (2021); CC BY 2.0; Wikimedia

[Commons](#)

Section Summary

- Align news and prices with care: date-only stamps push the first tradable return to $d + 2$
- Headline sentiment is strongly related to same-day returns, and barely to returns one can still trade
- Test signals with clustered errors, Newey–West t , costs and subperiods

What Is an AI Agent?

- ▣ An LLM in a loop: read inputs, decide, call tools, observe the result, repeat
 - ▶ Tools: data queries, code execution, news search, order entry through an API
 - ▶ Memory: notes and past decisions kept between steps
- ▣ Uses in finance today: research summaries, coding and data work, monitoring of news and filings
 - ▶ Fully autonomous trading by LLM agents is still rare and mostly experimental

Risks of Agents

- ▣ Model risks: invented facts, look-ahead in backtests, sensitivity to wording
- ▣ Manipulation: a text read by the agent can contain instructions (prompt injection), for example a fake press release
- ▣ System risks (IMF (2024)): many firms using similar models may trade alike, raising speed and correlation of moves in stress
 - ▶ The same report sees benefits too: faster processing of information and lower costs
- ▣ Controls: position limits, human approval for orders, logging of every step, tests after each model update

An Open Question: Can LLM-Generated Hypotheses Be Trusted?

- Ludwig & Mullainathan (2024): hypothesis generation as a statistical procedure: a model trained on data reveals a pattern, humans name it, fresh data test it
 - ▶ Their application: defendants' mugshots and judges' pretrial detention decisions
- Lopez-Lira, Tang & Zhu (2025): before its knowledge cutoff an LLM can recall outcomes, so a hypothesis it proposes about that period is not independent of the data that will test it
- Open: when an assistant proposes the signal, the prompt, the threshold and the holding period, what is the family of tests, and which inference stays valid? Agents that run the experiments themselves (Lu et al. (2024)) make the family larger and less visible
- Formally: $H_{0,j} : E[r_t^{(j)}] = 0$ for candidate strategies $j = 1, \dots, M$; control the FWER (Holm) or the FDR (Benjamini & Hochberg (1995))
 - ▶ M counts every variant the assistant proposed, including those discarded; trading agents are covered in AI Agents in Trading, here the agent is a research assistant

The Discovery Loop, Step by Step (1/2)

- **Literature:** "Find studies since 2023 that predict stock returns with LLM sentiment and test after the model's knowledge cutoff: DOI or arXiv id, cutoff date, test window, headline estimate."
 - ▶ Resolve every identifier; Semantic Scholar, Google Scholar, Elicit for the search; [Lopez-Lira & Tang \(2026\)](#) is the reference design
- **Hypotheses:** "Propose mechanisms by which headline sentiment could predict returns after d+1 and, for each, the cross-sectional pattern that would falsify it."
 - ▶ Under-reaction predicts drift concentrated in small, less-covered stocks; memorisation predicts an edge that disappears after the release date
- **Data and code:** "From ch15_news_daily.csv and data/market, reproduce the daily Qwen strategy of this chapter (enter at the close of d+1, hedge with SPY) and its Newey-West t."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot); the scores are saved, so no model needs to be re-run

The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed in advance: "Before any new run, write the full grid of variants you would test and the multiple-testing correction; all of them will be reported."
 - ▶ Placebo: scores shuffled across stocks within the same day; subperiods; costs of 5 bp per trade
- **Critique**: "Write a referee report: every way this strategy could reflect look-ahead, memorisation, selection or timestamps, with the test for each."
 - ▶ Then check each proposed test yourself: a critique can also contain invented results
- **Report**: the whole grid, the variant chosen before the results, model versions and release dates

What the Human Must Check

- **Memorisation:** the model knows what followed the headline
 - ▶ Test: news after the release of the weights; anonymised text (Glasserman & Lin (2023)); the memory test of this chapter
- **Forking paths:** prompt, threshold, horizon tuned on the test period
 - ▶ Test: a pre-registered grid; Holm, BH, or $|t| > 3$ (Harvey, Liu & Zhu (2016))
- **HARKing:** a plausible mechanism written after the result, with references that may not exist
 - ▶ Test: state the mechanism and its falsifier before the test; resolve every DOI
- **Timing:** date-only timestamps
 - ▶ Test: first tradable return on $d + 2$; the event study shows what is left on the tradable days
- **Agents that run code:** silent changes of data, windows or seeds
 - ▶ Test: read-only data, logged tool calls, the diff of every script before a result is accepted

Mini-Case: the Garden of Forking Paths

- A typical assistant proposal: "the Qwen signal works; refine it with a stronger threshold or a longer holding period". We test the whole grid on the saved scores
- Grid: 3 scores (LM, FinBERT, Qwen) $\times$ 3 thresholds $|s_{i,d}| > b$, $b \in \{0, 0.25, 0.5\}$, on the raw score $\times$ 4 holding days ($d + 2$ to $d + 5$) = 36 strategies
 - ▶ The base case replicates: 5.6 bp per day, $t = 2.26$
- $|t| > 1.96$ for 5 of 36 strategies, against 1.8 expected under the global null at 5%; 2 of them have the wrong sign
- Survivors: Holm (FWER 5%) 0, BH (FDR 10%) 0, $|t| > 3$ 0; the Bonferroni threshold is $|t| > 3.20$
- Reading: the refinements add no evidence; the best variant is the one we started from, itself chosen among three scores, on a period the model may remember

 `ai_discovery_case.py`

Your Turn: a Project Seed

- Question: does LLM headline sentiment predict returns beyond $d + 1$ on news published after the model's release, under a pre-registered grid?
- Replicate first: the Qwen strategy of this chapter, +5.64 bp per day ($t = +2.26$); then the post-cutoff design of [Lopez-Lira & Tang \(2026\)](#)
- Deliverables
 - ▶ The pre-registered grid and correction; dated headlines after the release of an open model; the test on all variants
 - ▶ An LLM-proposed mechanism, stated before the test with its falsifier; a referee-style critique and your reply
- Log each prompt that shaped a decision in AI_USE.md; each invented reference, memorised fact or wrong number caught in AI_ERRORS.md, with the test that caught it ([templates](#))

Pitfalls Checklist

- ▣ **Training overlap:** FinBERT scores 89.0% on its own training data, 72.5% on new data
- ▣ **Look-ahead:** an LLM may remember the market after the news; test after its release date
- ▣ **Timing:** date-only timestamps; trade no earlier than the close after the news is surely public
- ▣ **Snooping:** prompts, thresholds and models chosen after seeing the test results
- ▣ **Costs and crowding:** small daily edges vanish after a few basis points of costs

When to Use What

Situation	Choice	Why
Fast, transparent baseline	LM dictionary	no training, auditable
Short English financial texts	FinBERT	accurate, milliseconds per text
No labels, other languages, long documents	open LLM (7B+)	zero-shot, flexible
A few hundred labels from your domain	embeddings + logistic regression	cheap and adapted to the domain

- ▣ Whatever the method: evaluate out of sample, with intervals and paired tests

Related Research

- ▣ [Pele \(2024\)](#): sentiment analysis Quantlet of the BitMood project, IDA Quantlets repository
- ▣ [Pele \(2025\)](#): Dogecoin and speculative bubbles, studied with the log-periodic power law and sentiment analysis (Quantinar)
- ▣ [Pele et al. \(2023\)](#): statistical classification of crypto-assets: when do they behave like traditional assets?
- ▣ Further study: [Quantinar: Natural Language Processing](#), [Quantinar: DELTA](#), [Quantinar: LDA Extensions](#); textbook [Franke, Härdle & Hafner \(2019\)](#) for the statistics behind the tests

Key Takeaways

- Generic dictionaries misread finance; the LM dictionary is a transparent baseline
- FinBERT and 7B–14B open LLMs are statistically tied on financial headlines; a few hundred labels and a simple classifier compete with both
- LLMs remember market history: evaluate them only after their release date
- Headline sentiment explains same-day moves; with realistic timing and costs it does not beat the market in large stocks

References (1/7)

- ▣ Antweiler, W., Frank, M.Z. (2004). [Is all that talk just noise? The information content of internet stock message boards.](#) *Journal of Finance*, 59(3), 1259–1294.
- ▣ Araci, D. (2019). [FinBERT: financial sentiment analysis with pre-trained language models.](#) arXiv:1908.10063.
- ▣ Baker, M., Wurgler, J. (2006). [Investor sentiment and the cross-section of stock returns.](#) *Journal of Finance*, 61(4), 1645–1680.
- ▣ Benjamini, Y., Hochberg, Y. (1995). [Controlling the false discovery rate: a practical and powerful approach to multiple testing.](#) *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- ▣ Bollen, J., Mao, H., Zeng, X. (2011). [Twitter mood predicts the stock market.](#) *Journal of Computational Science*, 2(1), 1–8.
- ▣ Brown, T.B., Mann, B., Ryder, N., Subbiah, M., et al. (2020). [Language models are few-shot learners.](#) *NeurIPS*; arXiv:2005.14165.
- ▣ Devlin, J., Chang, M.-W., Lee, K., Toutanova, K. (2019). [BERT: pre-training of deep bidirectional Transformers for language understanding.](#) *NAACL*; arXiv:1810.04805.

References (2/7)

- ▣ Dong, Z., Fan, X., Peng, Z. (2024). [FNSPID: a comprehensive financial news dataset in time series](#). *Proceedings of the 30th ACM SIGKDD Conference*, 4918–4927.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ Gentzkow, M., Kelly, B., Taddy, M. (2019). [Text as data](#). *Journal of Economic Literature*, 57(3), 535–574.
- ▣ Glasserman, P., Lin, C. (2023). [Assessing look-ahead bias in stock return predictions generated by GPT sentiment analysis](#). arXiv:2309.17322.
- ▣ Gu, S., Kelly, B., Xiu, D. (2020). [Empirical asset pricing via machine learning](#). *Review of Financial Studies*, 33(5), 2223–2273.
- ▣ Harvey, C.R., Liu, Y., Zhu, H. (2016). [... and the cross-section of expected returns](#). *Review of Financial Studies*, 29(1), 5–68.
- ▣ Huang, A.H., Wang, H., Yang, Y. (2023). [FinBERT: a large language model for extracting information from financial text](#). *Contemporary Accounting Research*, 40(2), 806–841.

References (3/7)

- Hutto, C., Gilbert, E. (2014). [VADER: a parsimonious rule-based model for sentiment analysis of social media text](#). *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1), 216–225.
- IMF (2024). [Advances in artificial intelligence: implications for capital market activities](#). *Global Financial Stability Report*, October 2024, Chapter 3.
- Jegadeesh, N., Wu, D. (2013). [Word power: a new approach for content analysis](#). *Journal of Financial Economics*, 110(3), 712–729.
- Ke, Z.T., Kelly, B., Xiu, D. (2019). [Predicting returns with text data](#). NBER Working Paper 26186.
- Kearney, C., Liu, S. (2014). [Textual sentiment in finance: a survey of methods and models](#). *International Review of Financial Analysis*, 33, 171–185.
- Kim, A., Muhn, M., Nikolaev, V. (2023). [Bloated disclosures: can ChatGPT help investors process information?](#) arXiv:2306.10224.
- Lopez-Lira, A., Tang, Y. (2026). [Can ChatGPT forecast stock price movements? Return predictability and large language models](#). *Journal of Financial Economics*, 184, 104335.

References (4/7)

- ▣ Lopez-Lira, A., Tang, Y., Zhu, M. (2025). The memorization problem: can we trust LLMs' economic forecasts? [arXiv:2504.14765](#).
- ▣ Loughran, T., McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1), 35–65.
- ▣ Loughran, T., McDonald, B. (2016). Textual analysis in accounting and finance: a survey. *Journal of Accounting Research*, 54(4), 1187–1230.
- ▣ Lu, C., Lu, C., Lange, R.T., Foerster, J., Clune, J., Ha, D. (2024). The AI Scientist: towards fully automated open-ended scientific discovery. [arXiv:2408.06292](#).
- ▣ Ludwig, J., Mullainathan, S. (2024). Machine learning as a tool for hypothesis generation. *Quarterly Journal of Economics*, 139(2), 751–827.
- ▣ Malo, P., Sinha, A., Korhonen, P., Wallenius, J., Takala, P. (2014). Good debt or bad debt: detecting semantic orientations in economic texts. *Journal of the Association for Information Science and Technology*, 65(4), 782–796.
- ▣ Mikolov, T., Chen, K., Corrado, G., Dean, J. (2013). Efficient estimation of word representations in vector space. [arXiv:1301.3781](#).

References (5/7)

- ▣ Ouyang, L., Wu, J., Jiang, X., Almeida, D., et al. (2022). [Training language models to follow instructions with human feedback](#). *NeurIPS*; arXiv:2203.02155.
- ▣ Pele, D.T. (2024). [Sentiment Analysis](#) (Quantlet, BitMood project). IDA Quantlets Repository, Bucharest University of Economic Studies.
- ▣ Pele, D.T. (2025). [Dogecoin and speculative bubbles: a study using LPPL and sentiment analysis](#). Quantinar course, Bucharest University of Economic Studies.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#) *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Petersen, M.A. (2009). [Estimating standard errors in finance panel data sets: comparing approaches](#). *Review of Financial Studies*, 22(1), 435–480.
- ▣ Qwen Team (2024). [Qwen2.5 technical report](#). arXiv:2412.15115.
- ▣ Reimers, N., Gurevych, I. (2019). [Sentence-BERT: sentence embeddings using Siamese BERT-networks](#). *EMNLP*; arXiv:1908.10084.

References (6/7)

- ▣ Sarkar, S.K., Vafa, K. (2024). Lookahead bias in pretrained language models. SSRN Working Paper 4754678.
- ▣ Shannon, C.E. (1951). Prediction and entropy of printed English. *Bell System Technical Journal*, 30(1), 50–64.
- ▣ Shiller, R.J. (2017). Narrative economics. *American Economic Review*, 107(4), 967–1004.
- ▣ Tetlock, P.C. (2007). Giving content to investor sentiment: the role of media in the stock market. *Journal of Finance*, 62(3), 1139–1168.
- ▣ Tetlock, P.C., Saar-Tsechansky, M., Macskassy, S. (2008). More than words: quantifying language to measure firms' fundamentals. *Journal of Finance*, 63(3), 1437–1467.
- ▣ Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., et al. (2017). Attention is all you need. *NeurIPS*; arXiv:1706.03762.
- ▣ Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., Zhou, M. (2020). MiniLM: deep self-attention distillation for task-agnostic compression of pre-trained Transformers. *NeurIPS*; arXiv:2002.10957.

References (7/7)

- Wu, S., Irsoy, O., Lu, S., Dabrovolski, V., et al. (2023). [BloombergGPT: a large language model for finance](#). arXiv:2303.17564.